Supplement to *Males and Females (Machine) Learning to Express Emotions More Granularly*

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DATA 515: Advanced Applied Machine Learning

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Table S1

Baseline Model Accuracy and Parameters for DASS-21 Subscales as Predictors of Depression

Model	**Female**	**Male**	**Model**	**Female**	**Male**
Random Forest			**XGBoost**		
Accuracy (%)	98.36	97.58	*Accuracy (%)*	99.93	99.71
Best Parameters			*Best Parameters*		
N Estimators	250	500	Scale Position Weight	.44	.54
Minimum Samples Split	30	30	N Estimators	500	500
Maximum Features	.30	sqrt	Maximum Depth	3	3
Maximum Depth	30	30	Learning Rate	.50	.10
Criterion	Gini	Entropy	Columns Sampled by Tree	.30	.30
Class Weight	Balanced	Balanced			
AdaBoost			**LightGBM**		
Accuracy (%)	100	100	*Accuracy (%)*	99.78	98.73
Best Parameters			*Best Parameters*		
N Estimators	500	500	N Estimators	1000	250
Learning Rate	.50	.50	Maximum Depth	-1	10
Estimator Maximum Depth	2	2	Columns Sampled by Tree	.30	.50
Estimator Criterion	Entropy	Log Loss	Class Weight	Balanced	Balanced
Gradient Boosting			**CatBoost**		
Accuracy (%)	99.95	99.71	*Accuracy (%)*	99.95	99.94
Best Parameters			*Best Parameters*		
N Estimators	1000	1000	Learning Rate	.05	.05
Minimum Samples Split	30	30	Iterations	1000	1000
Maximum Features	.30	sqrt	Depth	3	3
Maximum Depth	3	3	Auto Class Weights	Balanced	Balanced
Histogram Gradient Boosting					
Accuracy (%)	99.78	98.91			
Best Parameters					
Minimum Samples Per Leaf	30	30			
Maximum Iterations	1000	500			
Maximum Depth	None	20			
Maximum Bins	128	255			
Learning Rate	.50	.50			

Table S2

Baseline Model Accuracy and Parameters for DASS-21 Subscales as Predictors of Anxiety

Model	Female	Male	Model	Female	Male
Random Forest			**XGBoost**		
Accuracy (%)	99.62	99.42	*Accuracy (%)*	99.93	100
Best Parameters			*Best Parameters*		
N Estimators	100	100	Scale Position Weight	.41	.73
Minimum Samples Split	30	30	N Estimators	500	500
Maximum Features	.50	.50	Maximum Depth	10	3
Maximum Depth	20	30	Learning Rate	.05	.05
Criterion	Gini	Gini	Columns Sampled by Tree	1	.50
Class Weight	Balanced	Balanced			
AdaBoost			**LightGBM**		
Accuracy (%)	99.98	100	*Accuracy (%)*	99.97	99.94
Best Parameters			*Best Parameters*		
N Estimators	1000	500	N Estimators	100	100
Learning Rate	.10	.50	Maximum Depth	10	20
Estimator Maximum Depth	2	2	Columns Sampled by Tree	1	1
Estimator Criterion	Entropy	Gini	Class Weight	Balanced	Balanced
Gradient Boosting			**CatBoost**		
Accuracy (%)	99.97	100	*Accuracy (%)*	99.98	100
Best Parameters			*Best Parameters*		
N Estimators	500	1000	Learning Rate	.50	.10
Minimum Samples Split	30	30	Iterations	1000	1000
Maximum Features	.30	.30	Depth	1	3
Maximum Depth	3	3	Auto Class Weights	Balanced	Balanced
Histogram Gradient Boosting					
Accuracy (%)	99.98	100			
Best Parameters					
Minimum Samples Per Leaf	30	30			
Maximum Iterations	250	1000			
Maximum Depth	20	20			
Maximum Bins	255	255			
Learning Rate	.10	.50			

Table S3

Baseline Model Accuracy and Parameters for DASS-21 Subscales as Predictors of Stress

Model	Female	Male
Random Forest		
Accuracy (%)	97.98	96.89
Best Parameters		
N Estimators	1000	100
Minimum Samples Split	30	30
Maximum Features	.50	.50
Maximum Depth	None	30
Criterion	Log Loss	Gini
Class Weight	Balanced	Balanced
AdaBoost		
Accuracy (%)	100	99.88
Best Parameters		
N Estimators	500	250
Learning Rate	.50	.50
Estimator Maximum Depth	1	2
Estimator Criterion	Log Loss	Log Loss
Gradient Boosting		
Accuracy (%)	99.98	99.83
Best Parameters		
N Estimators	1000	500
Minimum Samples Split	30	30
Maximum Features	.50	.50
Maximum Depth	3	3
Histogram Gradient Boosting		
Accuracy (%)	99.77	99.48
Best Parameters		
Minimum Samples Per Leaf	30	30
Maximum Iterations	250	250
Maximum Depth	30	20
Maximum Bins	255	255
Learning Rate	.50	.50

Model	Female	Male
XGBoost		
Accuracy (%)	99.77	99.77
Best Parameters		
Scale Position Weight	.63	1.12
N Estimators	250	500
Maximum Depth	30	3
Learning Rate	.50	.10
Columns Sampled by Tree	1	.30
LightGBM		
Accuracy (%)	99.73	99.48
Best Parameters		
N Estimators	1000	1000
Maximum Depth	-1	20
Columns Sampled by Tree	1	1
Class Weight	Balanced	Balanced
CatBoost		
Accuracy (%)	100	99.88
Best Parameters		
Learning Rate	.50	.50
Iterations	500	250
Depth	1	1
Auto Class Weights	Balanced	Balanced

Table S4

Baseline Model Accuracy and Parameters for Demographic Variables as Predictors of Depression

Model	Female	Male	Model	Female	Male
Random Forest			**XGBoost**		
Accuracy (%)	72.81	73.96	*Accuracy (%)*	73.06	74.08
Best Parameters			*Best Parameters*		
N Estimators	500	250	Scale Position Weight	.44	.54
Minimum Samples Split	30	30	N Estimators	500	1000
Maximum Features	.50	sqrt	Maximum Depth	20	10
Maximum Depth	20	30	Learning Rate	.05	.01
Criterion	Gini	Log Loss	Columns Sampled by Tree	.50	.30
Class Weight	Balanced	Balanced			
AdaBoost			**LightGBM**		
Accuracy (%)	76.11	75.40	*Accuracy (%)*	72.35	72.98
Best Parameters			*Best Parameters*		
N Estimators	500	1000	N Estimators	500	250
Learning Rate	.10	.10	Maximum Depth	30	10
Estimator Maximum Depth	3	3	Columns Sampled by Tree	.50	.50
Estimator Criterion	Log Loss	Gini	Class Weight	Balanced	Balanced
Gradient Boosting			**CatBoost**		
Accuracy (%)	76.16	75.69	*Accuracy (%)*	72.53	73.16
Best Parameters			*Best Parameters*		
N Estimators	100	500	Learning Rate	.10	.05
Minimum Samples Split	30	30	Iterations	1000	250
Maximum Features	.30	.10	Depth	6	6
Maximum Depth	3	3	Auto Class Weights	Balanced	Balanced
Histogram Gradient Boosting					
Accuracy (%)	76.28	76.04			
Best Parameters					
Minimum Samples Per Leaf	30	30			
Maximum Iterations	100	250			
Maximum Depth	10	20			
Maximum Bins	255	255			
Learning Rate	.05	.01			

Table S5

Baseline Model Accuracy and Parameters for Demographic Variables as Predictors of Anxiety

Model	Female	Male	Model	Female	Male
Random Forest			**XGBoost**		
Accuracy (%)	70.84	70.33	*Accuracy (%)*	72.17	70.33
Best Parameters			*Best Parameters*		
N Estimators	1000	250	Scale Position Weight	.41	.73
Minimum Samples Split	30	30	N Estimators	500	1000
Maximum Features	.50	.50	Maximum Depth	20	10
Maximum Depth	30	None	Learning Rate	.05	.01
Criterion	Entropy	Entropy	Columns Sampled by Tree	.50	.30
Class Weight	Balanced	Balanced			
AdaBoost			**LightGBM**		
Accuracy (%)	75.55	71.95	*Accuracy (%)*	70.32	70.10
Best Parameters			*Best Parameters*		
N Estimators	100	100	N Estimators	1000	100
Learning Rate	.05	.10	Maximum Depth	10	30
Estimator Maximum Depth	3	3	Columns Sampled by Tree	.50	.50
Estimator Criterion	Entropy	Entropy	Class Weight	Balanced	Balanced
Gradient Boosting			**CatBoost**		
Accuracy (%)	75.43	72.47	*Accuracy (%)*	69.86	71.54
Best Parameters			*Best Parameters*		
N Estimators	250	250	Learning Rate	.10	.05
Minimum Samples Split	30	30	Iterations	1000	500
Maximum Features	.10	.30	Depth	6	1
Maximum Depth	3	3	Auto Class Weights	Balanced	Balanced
Histogram Gradient Boosting					
Accuracy (%)	75.73	71.14			
Best Parameters					
Minimum Samples Per Leaf	30	30			
Maximum Iterations	250	250			
Maximum Depth	None	20			
Maximum Bins	255	255			
Learning Rate	.05	.01			

Table S6

Baseline Model Accuracy and Parameters for Demographic Variables as Predictors of Stress

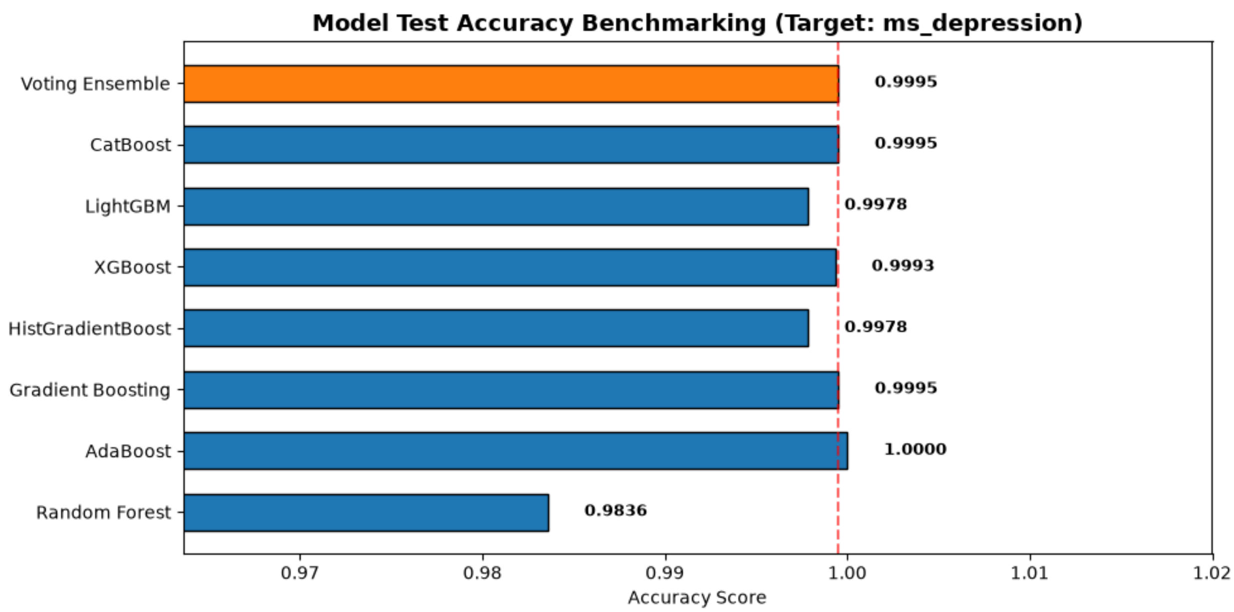
Model	Female	Male
Random Forest		
Accuracy (%)	73.39	72.98
Best Parameters		
N Estimators	100	1000
Minimum Samples Split	30	30
Maximum Features	sqrt	sqrt
Maximum Depth	None	30
Criterion	Gini	Entropy
Class Weight	Balanced	Balanced
AdaBoost		
Accuracy (%)	74.42	73.50
Best Parameters		
N Estimators	500	1000
Learning Rate	.05	.10
Estimator Maximum Depth	2	3
Estimator Criterion	Entropy	Log Loss
Gradient Boosting		
Accuracy (%)	74.93	73.44
Best Parameters		
N Estimators	500	100
Minimum Samples Split	30	30
Maximum Features	.10	.30
Maximum Depth	3	3
Histogram Gradient Boosting		
Accuracy (%)	74.85	73.50
Best Parameters		
Minimum Samples Per Leaf	30	30
Maximum Iterations	1000	100
Maximum Depth	10	30
Maximum Bins	255	255
Learning Rate	.01	.01

Model	Female	Male
XGBoost		
Accuracy (%)	73.36	73.68
Best Parameters		
Scale Position Weight	.63	1.12
N Estimators	100	100
Maximum Depth	3	30
Learning Rate	.01	255
Columns Sampled by Tree	1	.01
LightGBM		
Accuracy (%)	73.33	72.35
Best Parameters		
N Estimators	250	100
Maximum Depth	20	-1
Columns Sampled by Tree	.50	.30
Class Weight	Balanced	Balanced
CatBoost		
Accuracy (%)	73.36	73.56
Best Parameters		
Learning Rate	.01	.50
Iterations	100	100
Depth	3	1
Auto Class Weights	Balanced	Balanced

Figure S1

Model Benchmark Reports for DASS-21 Subscale Predictors of Depression

Females



Males

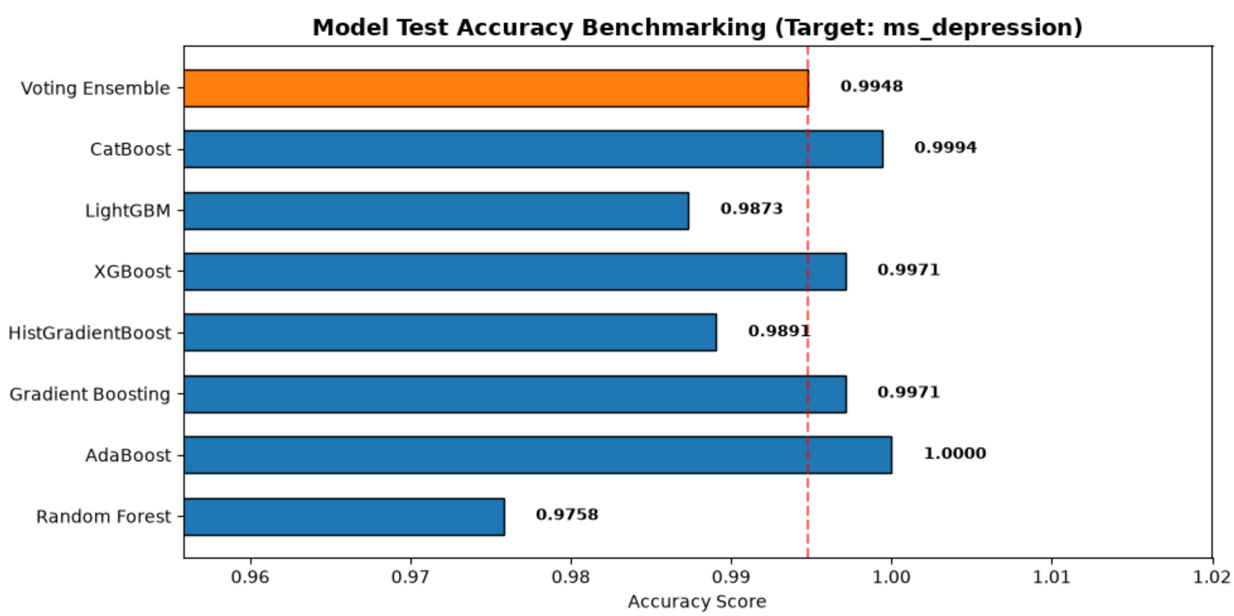
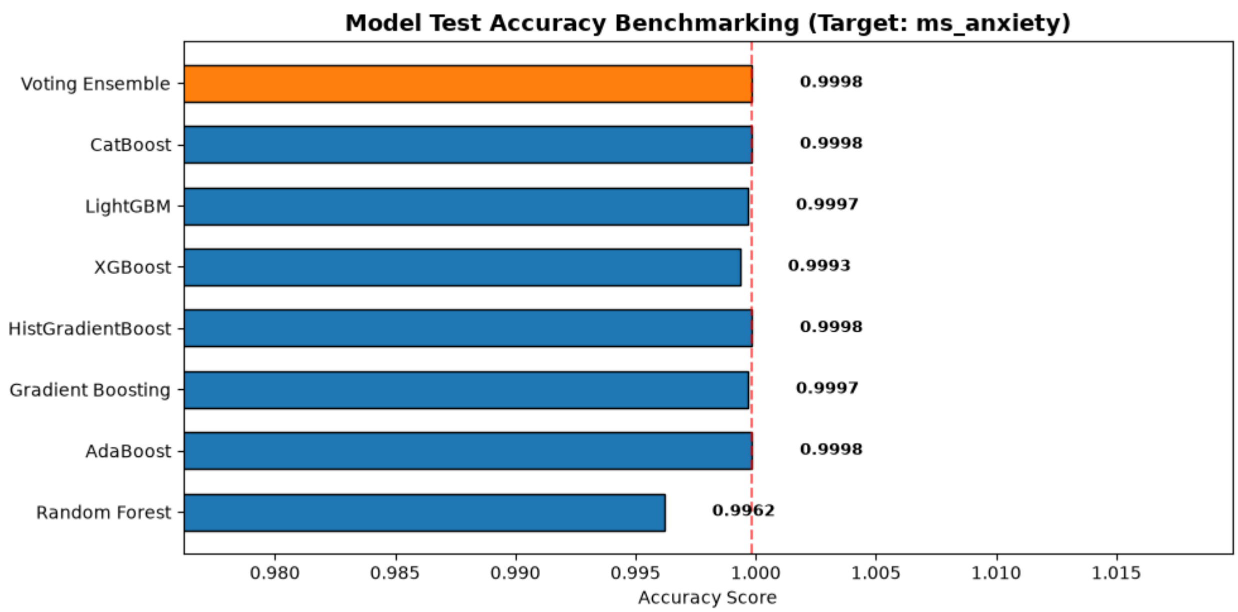


Figure S2

Model Benchmark Reports for DASS-21 Subscale Predictors of Anxiety

Females



Males

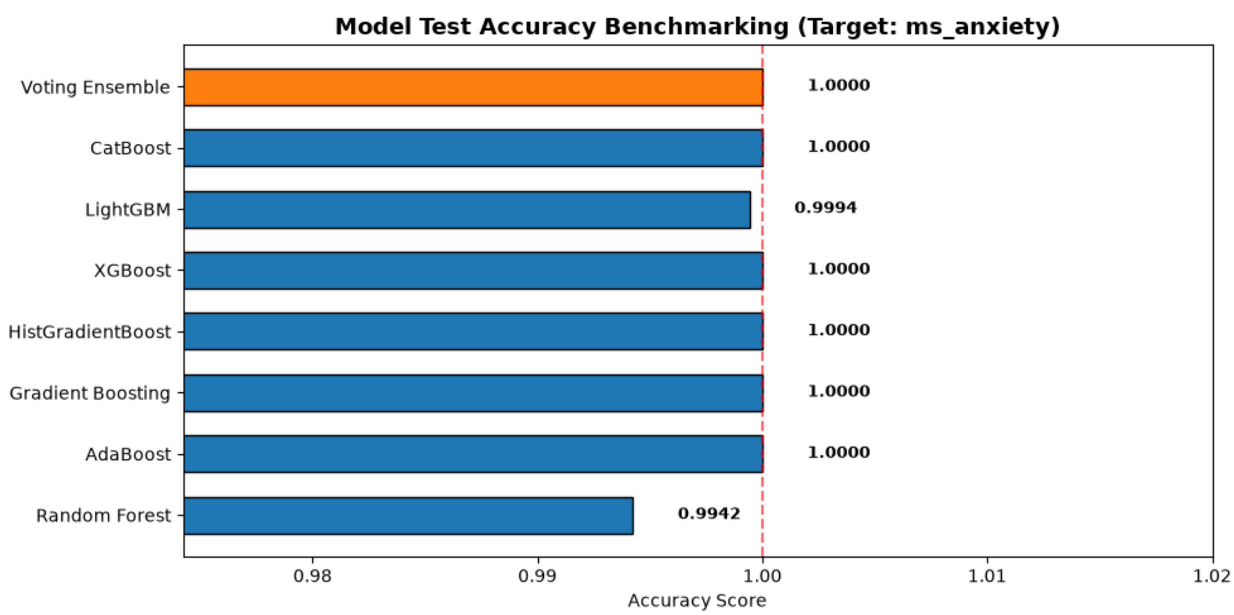
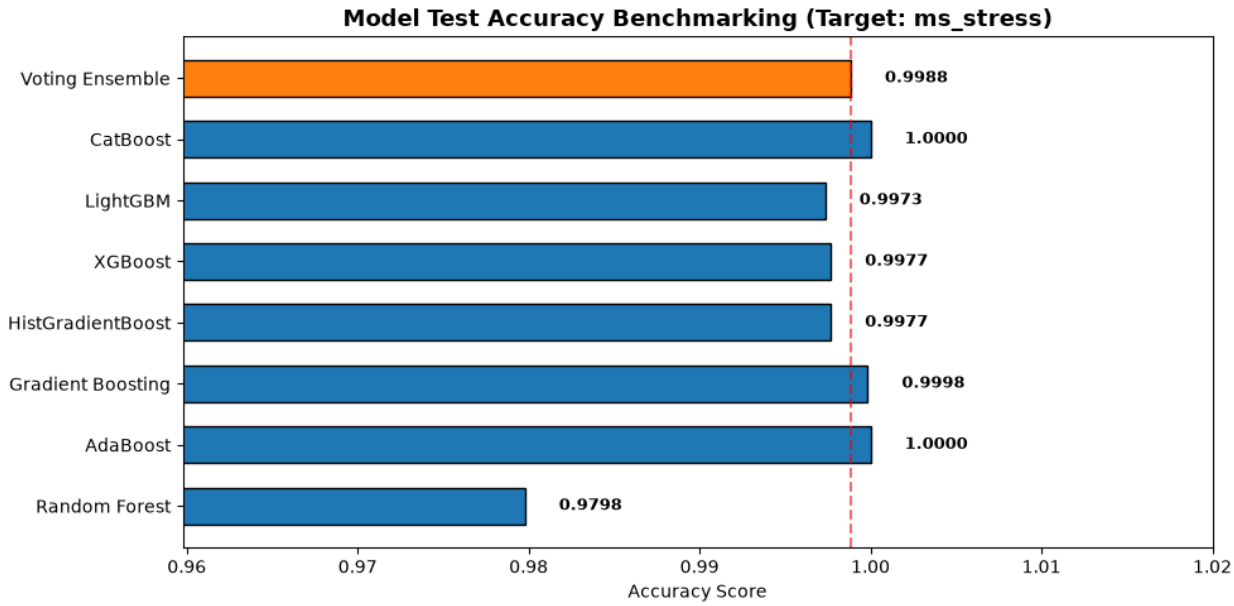


Figure S3

Model Benchmark Reports for DASS-21 Subscale Predictors of Stress

Females



Males

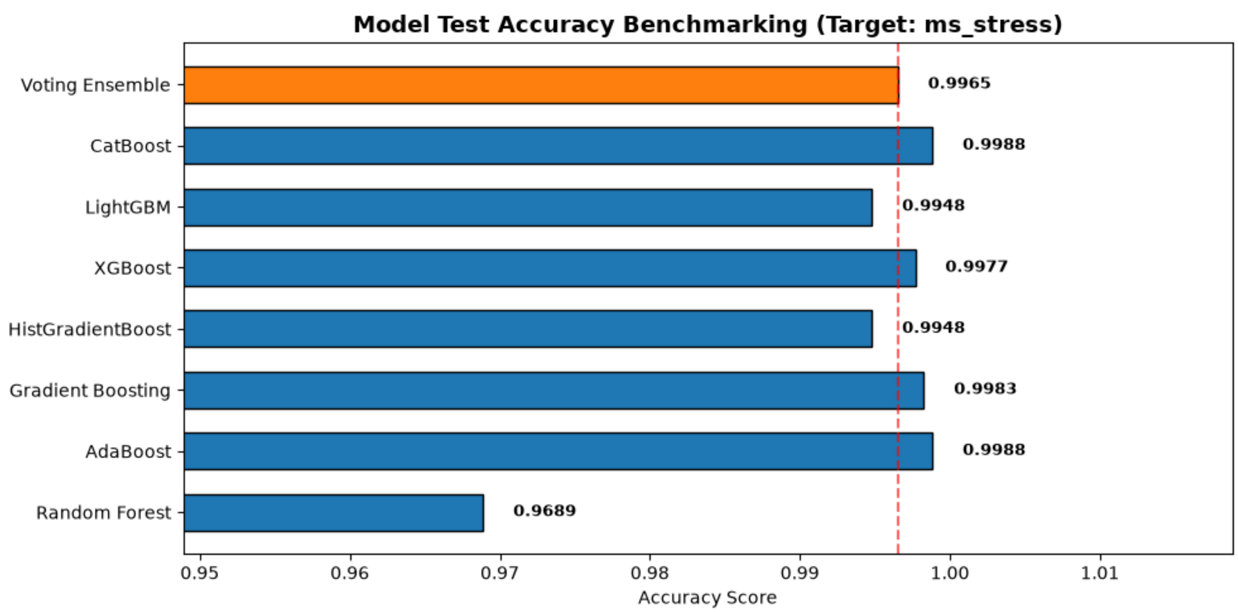
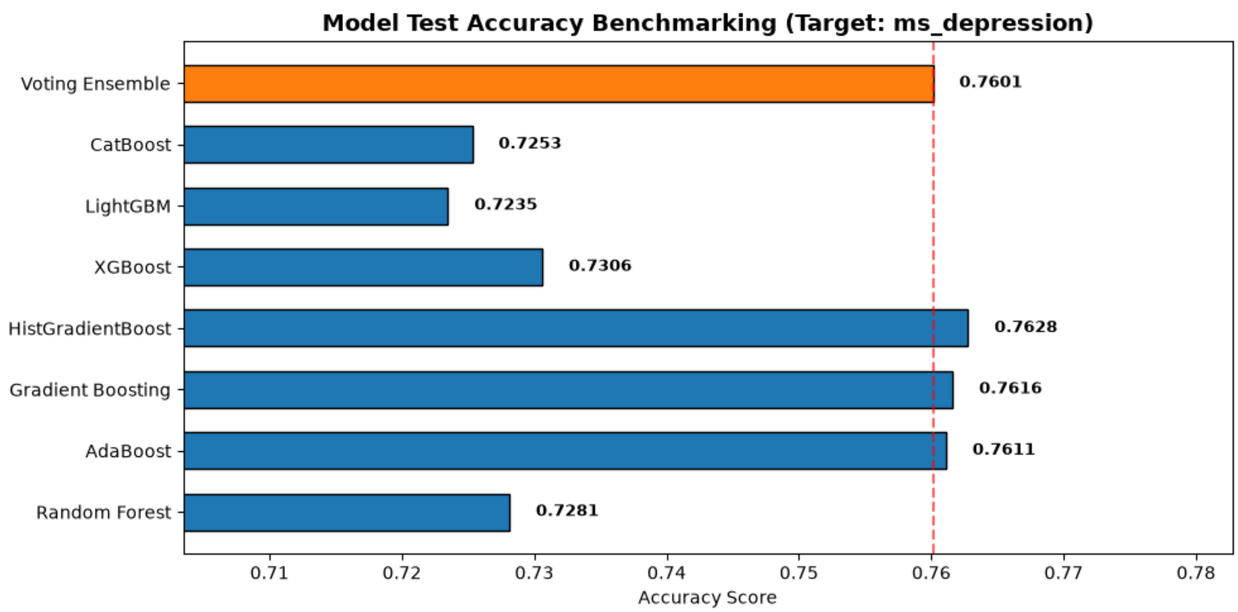


Figure S4

Model Benchmark Reports for Demographic Predictors of Depression

Females



Males

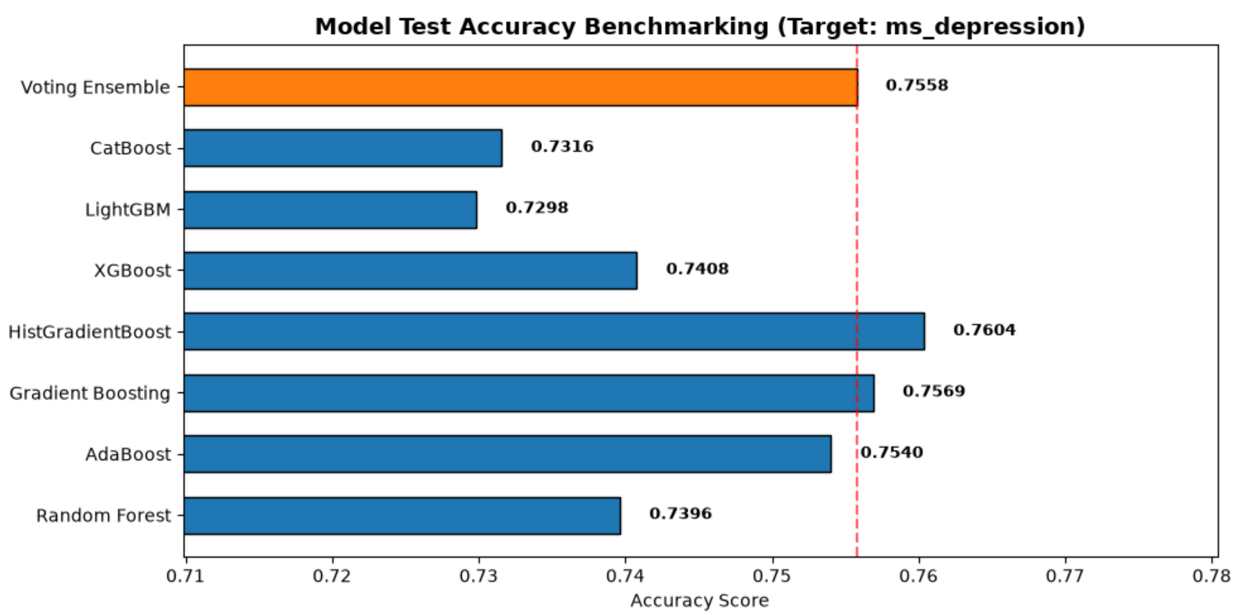
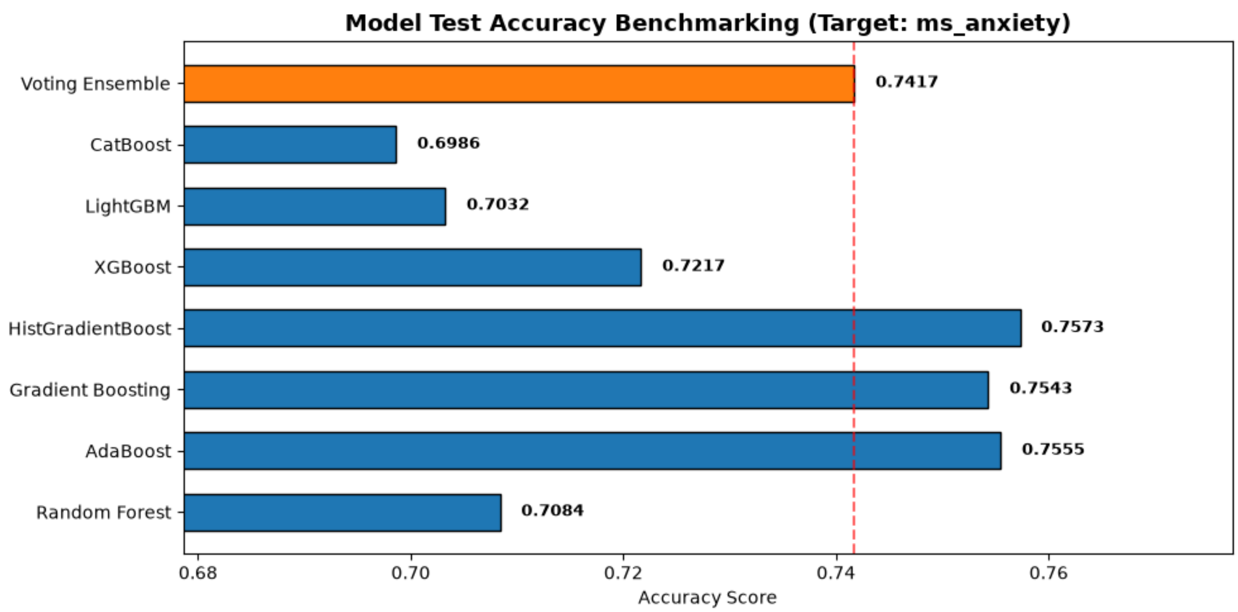


Figure S5

Model Benchmark Reports for Demographic Predictors of Anxiety

Females



Males

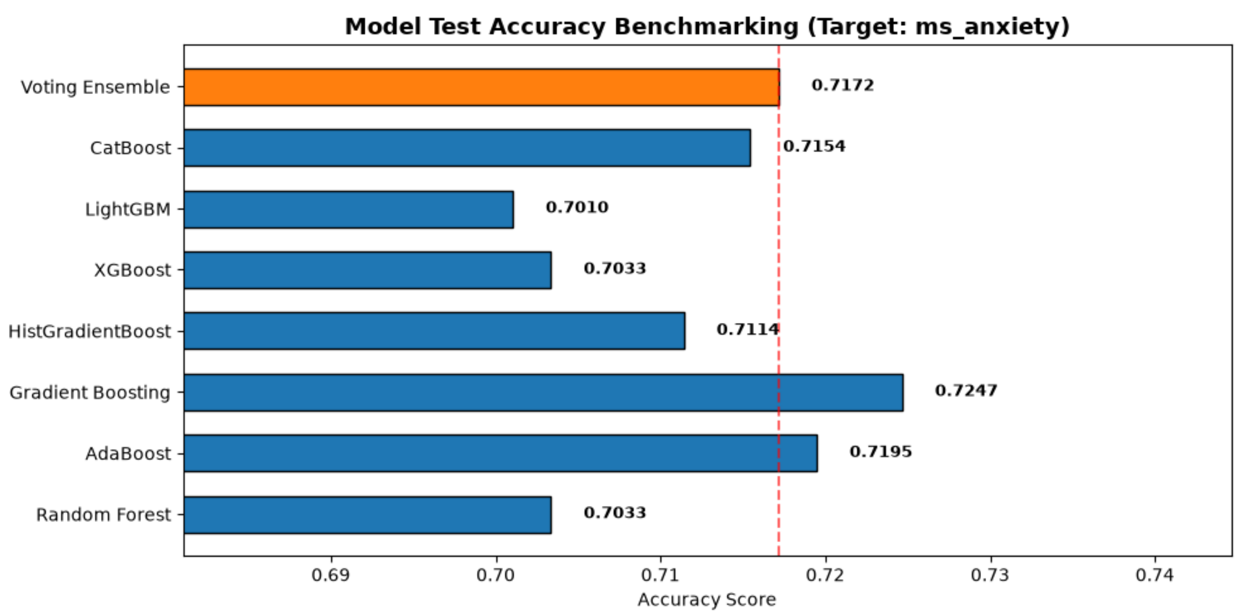
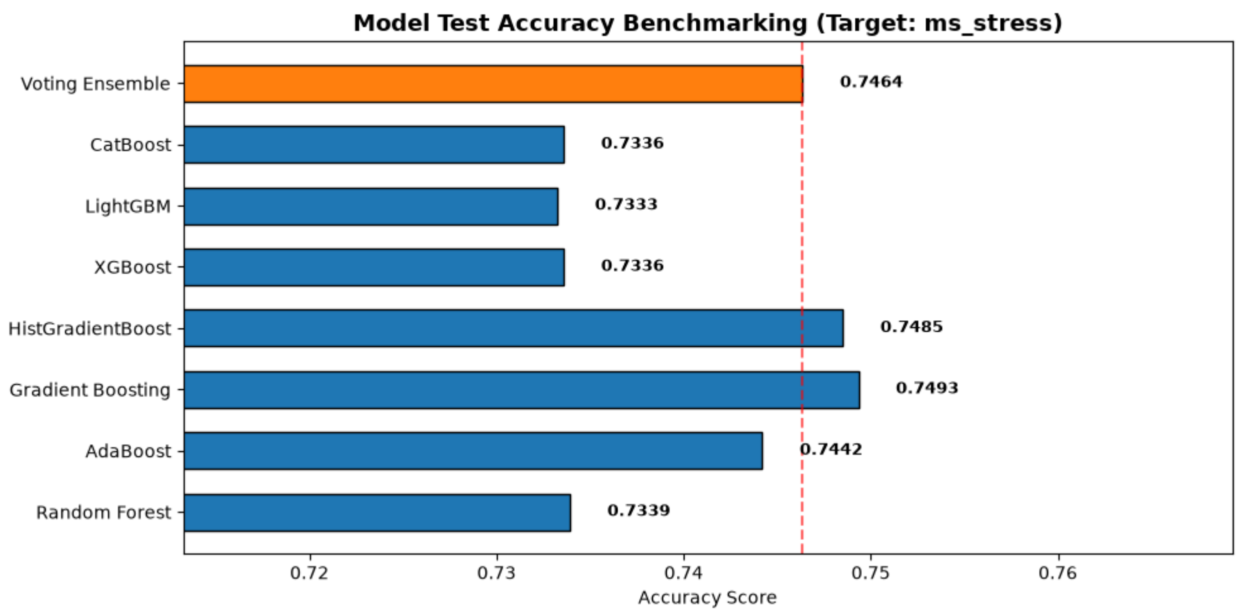


Figure S6

Model Benchmark Reports for Demographic Predictors of Stress

Females



Males

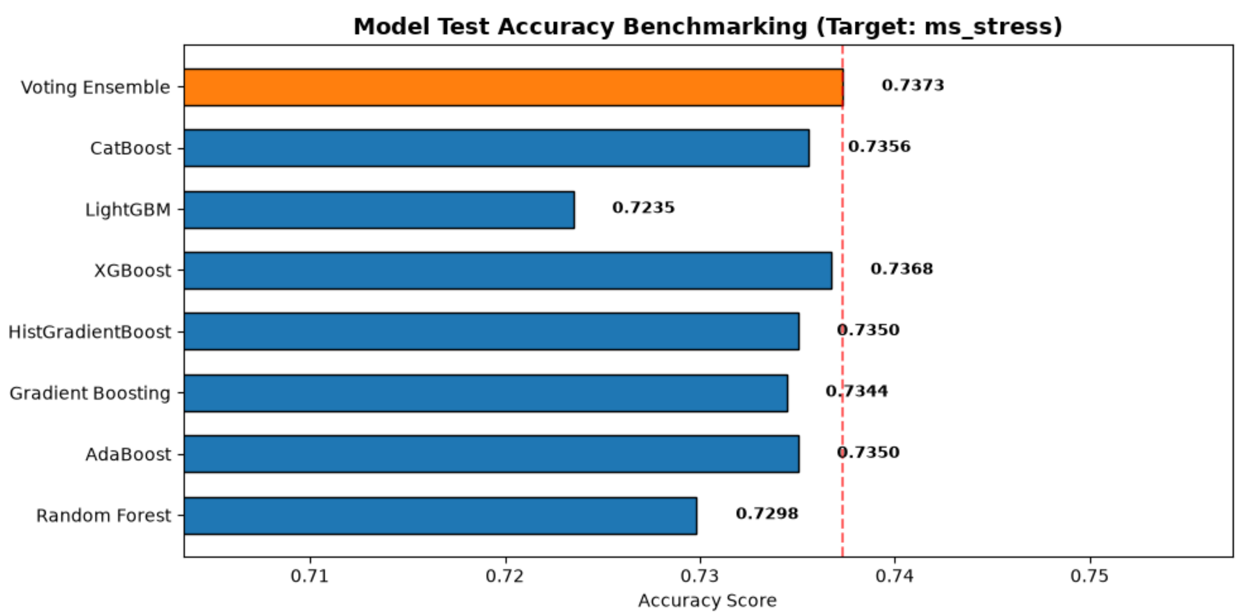
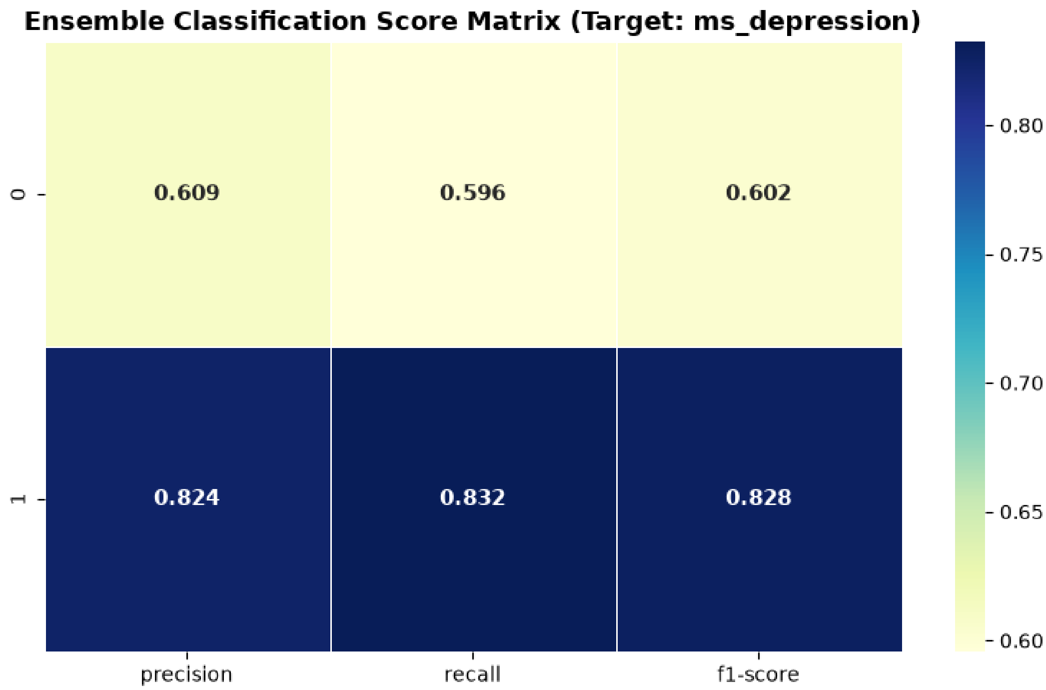


Figure S7

Model Confusion Matrices for Demographic Predictors of Depression

Females



Males

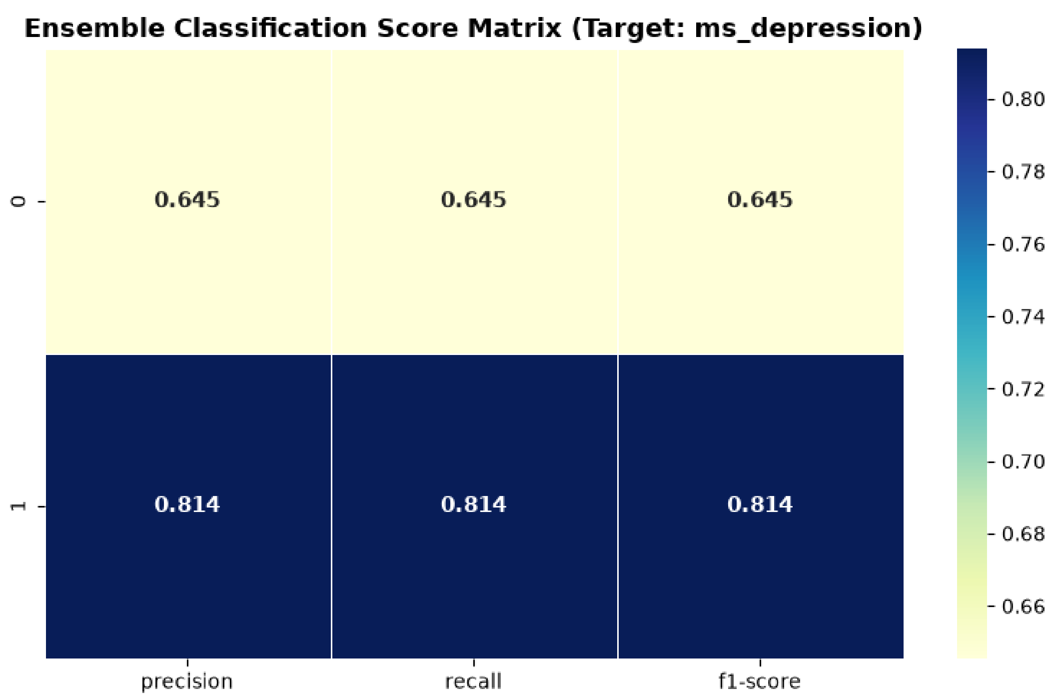
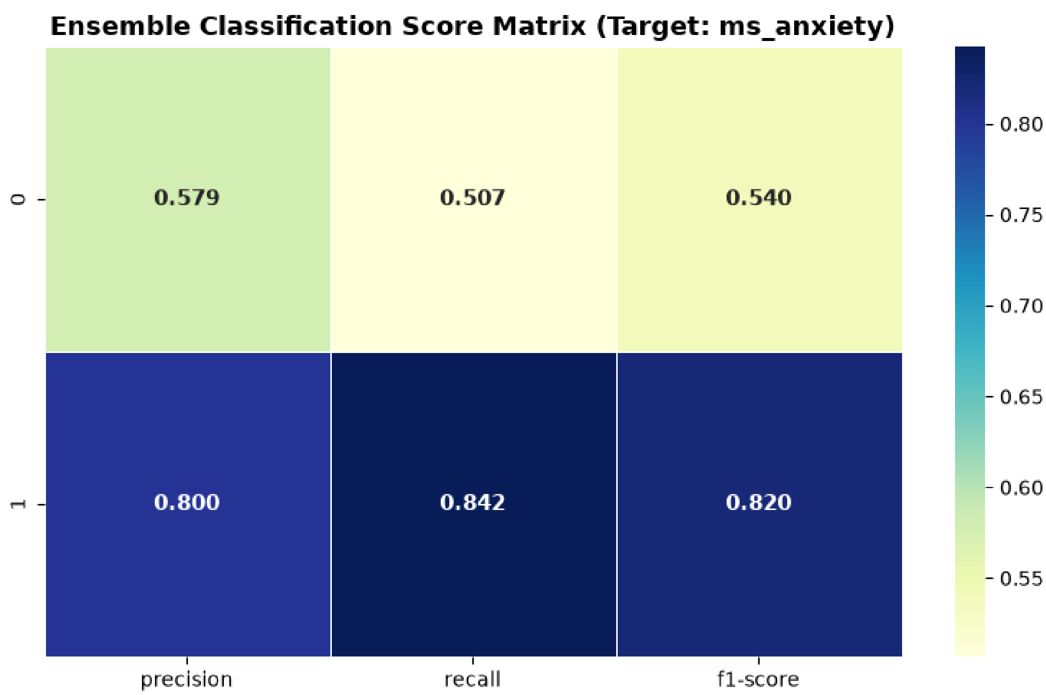


Figure S8

Model Confusion Matrices for Demographic Predictors of Anxiety

Females



Males

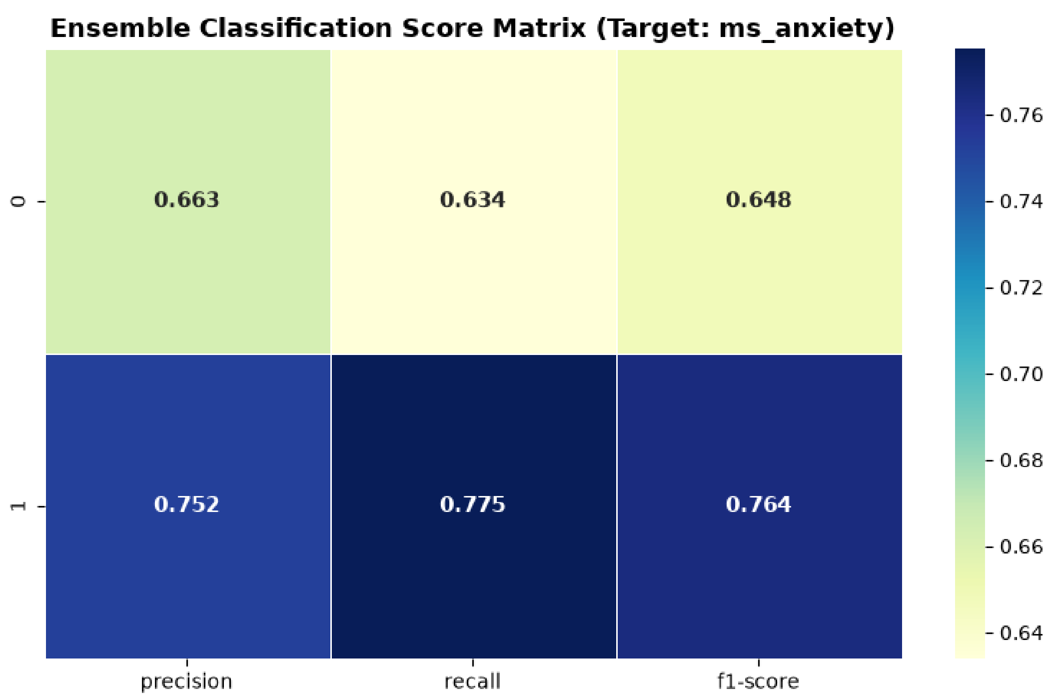
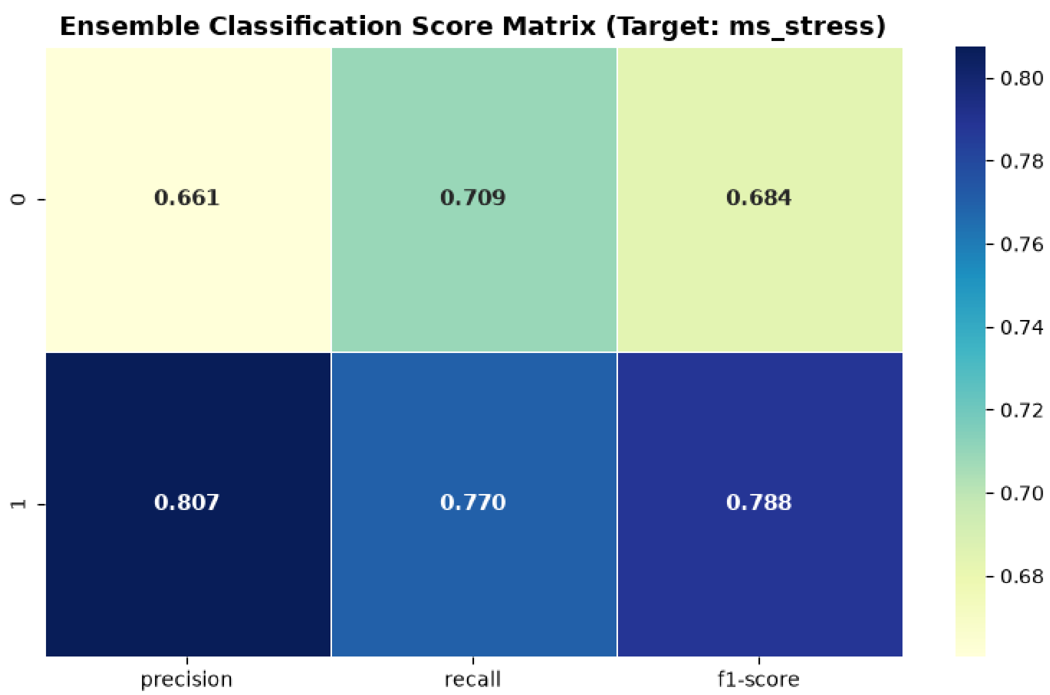


Figure S9

Model Confusion Matrices for Demographic Predictors of Stress

Females



Males